Lidar Introduction

1. Preface

Lidar is a remote sensing technology that uses laser beams to detect target positions, velocities, and other characteristics. It has the advantages of high measurement resolution, strong penetration ability, strong anti-interference ability, and strong anti-stealth ability.

According to the measurement principle, Lidar can be divided into three types: triangulation Lidar, pulse Lidar, and coherent Lidar. The Lidar equipped with MentorPi belongs to the pulse Lidar.

2. Lidar Principle

A Lidar system consists of a laser emission system, a scanning system, a laser reception system, and a signal processing system.

First, the laser emission system emits a detection signal (laser beam) towards the target, while the scanning system is responsible for scanning the plane to obtain the plane map information.

Next, the laser reception system receives the laser reflected from the target object and generates a reception signal.

Finally, the signal processing system processes the reception signal to obtain the characteristics of the target surface, physical properties, and other features, such as target location, height, and speed. Then, complete the construction of the object model.

3. Lidar Parameter

The specifications of the Lidar used by MentorPi are as follows:

3.1 Electrical and Mechanical Parameter

Parameter	Unit	Minimum value	Typical value	Maximum value	Note
Input voltage	V	4.5V	5V	5V	
PWM control frequency KHz	20	30	50	Square signal	
PWM high level	V	3	3	5	
PWM low level	V	-0.3	0	0.5	
PWM duty ratio	%	0	40	100	40% duty cycle with scanning frequency of 10Hz
Start-up current	mA	-	300	-	
Working current	mA	-	180	-	
Size	mm	54*46.29*34.8 (length*width*height)			
Wight	g	-	47	-	Connection cables are not included
Communication interface	-	UART@230400			

UART high level	V	2.9	3.3	3.5	
UART low level	V	-0.3	0	0.4	
Drive motor	-	BLDC	BLDC		
Working temperature	℃	-10	25	40	
Storage temperature	℃	-30	25	70	

3.2 Optical Parameter

Parameter	Unit	Minimum value	Typical value	Maximum value	Note
Laser wavelength	nm	895	905	915	Infrared wavelength
Laser power	W	-	25	-	Peak power of laser diode; the actual power used is much lower than this value
Laser pulse width	ns	-	1	-	
Laser safety class	-	IEC-60825 Class 1			
Pitch angle	°	0	0.5	2	

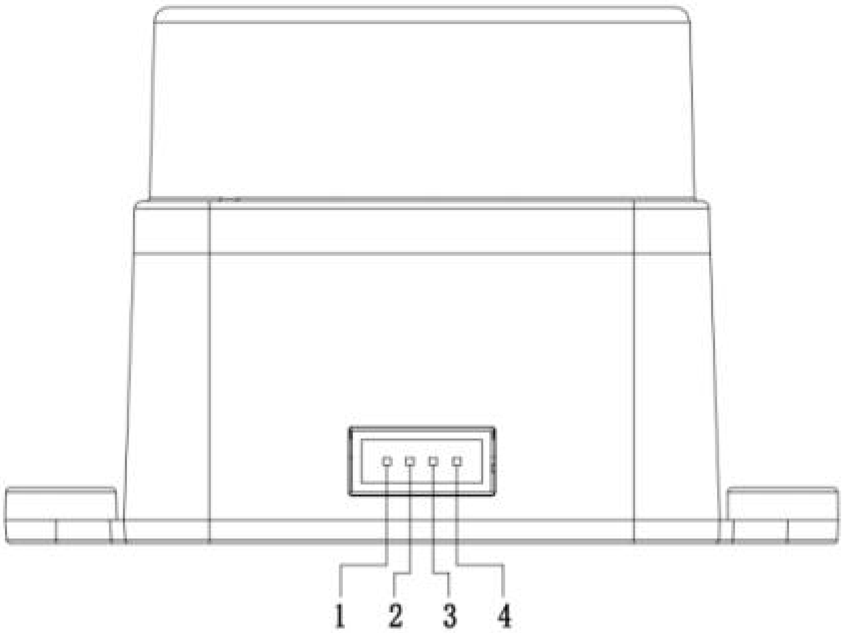
3.3 Performance Parameter

Parameter	Unit	Minimum value	Typical value	Maximum value	Note
Measuring range	m	0.02	-	12	70% target reflection coefficient
Scanning frequency	Hz	5	10	13	External PWM speed control
Ranging frequency	Hz	-	4500	-	Fixed frequency
Ranging accuracy	mm	-	-	-	When the ranging is less than 0.3m, there is output data. The trend of distance

					measurement data changes consistent with the trend of actual distance changes.
	mm	- 45	-	45	When the measuring range is between 300mm and 12000mm, the average of 100 measurements (70% diffuse surface).
Ranging standard deviation	mm	-	10	-	When the measuring range is between 300mm and 12000mm
Measurement resolution	mm	-	15	-	
Angle error	°	-	-	2	
Angle resolution	°	-	1	-	
Anti-environmental light interference	KLux	-	-	30	
Service life of the device	h	10000	-	-	

3.4 Interface Instruction and Communication Protocol

LD19 uses ZH1.5T-4P 1.5mm connector to connect with external systems for power supply and data reception. The specific interface definition and parameter are shown in the table below:



No.	Signal	Type	Description	Minimum value	Typical value	Maximum value
1	Tx	Output	Lidar data output	0V	3.3V	3.5V
2	PWM	Input	Motor control signal	0V	-	3.3V
3	GND	Power supply	Power supply negative terminal	-	0V	-
4	P5V	Power supply	Power supply positive terminal	4.5V	5V	5.5V

3.5 Serial Specification

You can connect the Lidar to external systems through its physical interface. By following the communication protocol of the system, you can obtain real-time point cloud data, device information and device status from the laser scanning, and can also set the device working mode.

Baud Rate	Data Length	Stop Bit	Parity Bit	Flow Control
230400	8 Bits	1	None	None